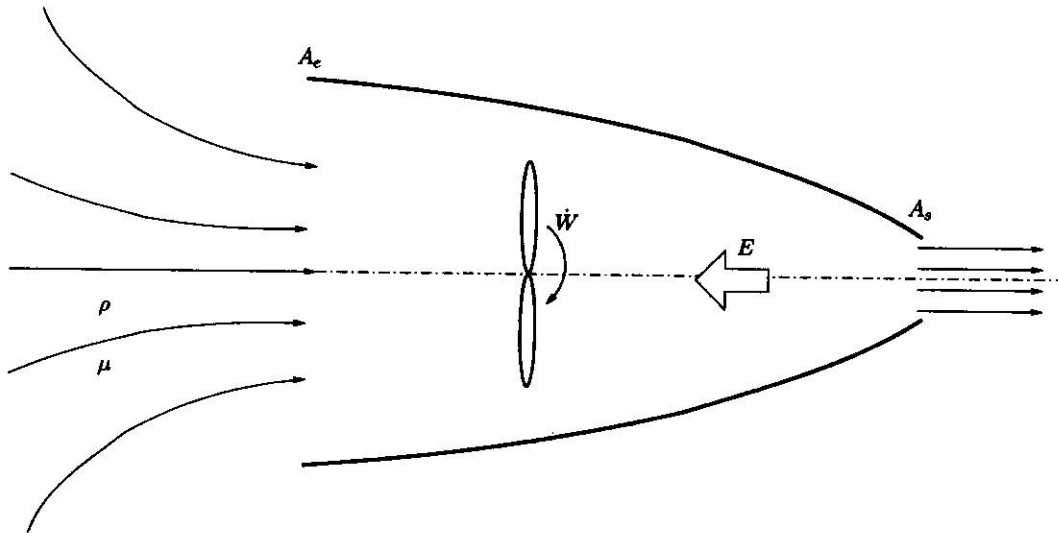


The submarine propulsion device schematically represented in the figure consists of a nozzle, with inlet and outlet crosssection area A_e and A_s , and a fan located inside, which imparts a rate of work per unit time (power) $\dot{W}$ to the circulating water stream, whose density and viscosity are ρ and μ , respectively. To characterize the thrust E produced when the submarine is at rest and at sufficient depth for gravity to have a negligible effect, please follow these steps:

1. By application of the Π theorem, reduce the parametric dependence of the thrust E .
2. Simplify the result assuming that the viscosity has a negligible effect, **justifying** the simplification. Maintain this assumption in addressing the next question.
3. An experimental test is conducted with a geometrically similar model with scale 1/10. It is found that, for a fan power $\dot{W}_e = 100$ Watts the resulting thrust is $E_e = 125$ N. If a thrust $E_p = 12500$ N is needed for the real prototype, determine the power $\dot{W}$ necessary to drive the fan.



$$1) E = E(\dot{W}, \rho, \mu, A_e, A_s) \rightarrow \frac{E}{\rho^{1/3} \dot{W}^{2/3} A_s^{1/3}} = f\left(\frac{A_e}{A_s}, \frac{\rho^{2/3} A_s^{1/6} \dot{W}^{1/3}}{\mu}\right)$$

$$2) \text{ IF } \frac{\rho^{2/3} A_s^{1/6} \dot{W}^{1/3}}{\mu} \gg 1 \rightarrow \text{VISCOUSITY IS NEGLIGIBLE} \rightarrow \frac{E}{\rho^{1/3} \dot{W}^{2/3} A_s^{1/3}} = f\left(\frac{A_e}{A_s}\right)$$

$$3) \frac{E}{\rho^{1/3} \dot{W}^{2/3} A_s^{1/3}} = \frac{E_e}{\rho^{1/3} \dot{W}_e^{2/3} A_{se}^{1/3}} \rightarrow \left[\dot{W} = \left(\frac{E}{E_e} \right)^{3/2} \left(\frac{A_{se}}{A_s} \right)^{1/2} \dot{W}_e \right] = \overset{100 \text{ Watts}}{10^4 \text{ Watts}}$$

$\downarrow \frac{3}{2}$ $\downarrow \frac{1}{2}$
 100 $\frac{1}{10}$
 "10000"